Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Recombinant production of high-titer retinoic acid from recombinant Yarrowia lipolytica			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		BioMedical	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	Pharmaceutical Intermediates	Industrial Biotechnology	Microbial and Enzyme Catalysts
Concept of Proposed Technology	Retinoic acid is a high-value active ingredient widely used in prescription dermatology, oncology, and cosmetic formulations for the treatment of acne, psoriasis, photoaging, and certain cancers. The global market for retinoids exceeds **$1–2 billion annually**, driven by sustained demand for topical and systemic therapies as well as premium cosmeceutical products. Despite this demand, industrial retinoic acid production remains dominated by multi-step chemical synthesis from petrochemical or plant-derived intermediates, resulting in energy-intensive processes, hazardous waste generation, and fragile supply chains subject to agricultural and geopolitical risk. Increasing regulatory scrutiny, sustainability mandates, and consumer demand for clean-label, environmentally responsible products are placing pressure on manufacturers to adopt greener production methods. A bio-based microbial manufacturing route for retinoic acid offers a compelling solution by enabling scalable, animal-free, and sustainable production directly from renewable feedstocks such as glucose.			

Development of such a platform would address critical supply-chain vulnerabilities, reduce environmental impact, and create a competitive domestic manufacturing pathway for a strategically important pharmaceutical and cosmetic ingredient.

Technology Platform: The Koffas Lab at Rensselaer Polytechnic Institute has developed a high-performance metabolic engineering platform for the overproduction of β-carotene in the oleaginous yeast *Yarrowia lipolytica*, establishing a robust precursor supply for downstream retinoic acid biosynthesis. β-Carotene titers were enhanced through a combination of rational pathway amplification and host lipid metabolism engineering. Specifically, two chromosomal copies of the *ACC1* and *DGA* genes were introduced to increase acetyl-CoA availability and lipid accumulation, thereby expanding intracellular storage capacity for the hydrophobic carotenoid product. In parallel, key genes of the native mevalonate pathway were overexpressed via single-copy chromosomal integration to boost isoprenoid precursor flux. To further drive product formation, six copies of the β-carotene biosynthetic genes (*GGPPsa*, *CarRP*, and *CarB*) were integrated into the genome.

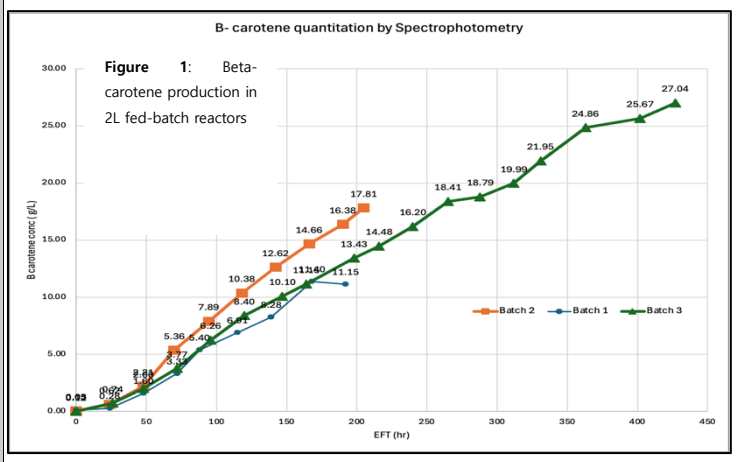

Figure 1: Beta-carotene production in 2L fed-batch reactors

This integrated strain engineering strategy, combined with systematic shake-flask, bioreactor, and fed-batch fermentation optimization, resulted in β-carotene titers exceeding **20 g/L (Figure 1, Figure 2)**, placing this platform among the highest reported microbial production systems. The strain exhibits stable growth, efficient intracellular carotenoid accumulation, and reproducible product quantification by spectrophotometry and HPLC. This work establishes a scalable, GRAS-compatible, and industrially relevant β-carotene production platform that directly enables the enzymatic conversion of β-carotene to retinoic acid in a fully bio-based process.

Figure 2 A 2L fed-batch fermentation for beta-carotene production

Academic Team: Prof. Mattheos Koffas is the Dorothy and Fred Chau ’71 Endowed Professor in the Department of Chemical and Biological Engineering at Rensselaer Polytechnic Institute. His research expertise lies in synthetic biology and metabolic engineering, with a strong focus on developing high-titer microbial production platforms for plant natural products, terpenoids, carotenoids, and complex bioactive molecules. Prof. Koffas has extensive experience in strain engineering, pathway optimization, and fermentation scale-up, providing the scientific foundation for translating β-carotene overproduction into a sustainable, glucose-to-retinoic-acid biomanufacturing platform.

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Rationale for Joint R&D Project	We are actively seeking industry collaborators with interests in cosmetics, personal care, and biomedical ingredients produced via advanced microbial fermentation. Strategic co-development with an industry partner will enable the definition of realistic, market-driven performance, cost, and scale-up targets, ensuring that the resulting technology aligns with commercial requirements. In addition, collaboration with industry will facilitate early validation of product performance in relevant formulations and applications, as well as evaluation by downstream customers. These partnerships will accelerate translation of the technology from laboratory-scale demonstration to commercially viable biomanufacturing platforms.
Key Words	1. Metabolic engineering of *Yarrowia lipolytica* 2. High-titer β-carotene fermentation platform 3. Bio-based retinoid acid production 4. Scalable, sustainable biomanufacturing 5. Mevalonate pathway amplification

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The overall objective of this project is to develop a scalable, bio-based microbial manufacturing platform for the production of retinoic acid from renewable carbon sources. The technology centers on advanced metabolic engineering of the oleaginous yeast *Yarrowia lipolytica* to enable high-titer synthesis of β-carotene and its efficient downstream enzymatic conversion to retinoic acid. Specific goals include: (1) engineering recombinant yeast strains with enhanced flux through acetyl-CoA, the mevalonate pathway, and the β-carotene biosynthetic pathway to maximize intracellular precursor accumulation; (2) introducing and functionally integrating a multi-step enzymatic module for the quantitative conversion of β-carotene to retinoic acid while maintaining intracellular redox balance; and (3) optimizing fermentation conditions and downstream analytical workflows to achieve reproducible, high-purity retinoic acid production at industrially relevant titers. Collectively, these technologies will establish a robust, animal-free, and sustainable fermentation platform suitable for translation to pharmaceutical and cosmetic ingredient manufacturing.			
R&D Target Specifications	Target performance specifications for this R&D project will be established and refined in collaboration with an industry partner to ensure commercial relevance. Key specifications will include product quality, process performance, and manufacturing productivity metrics. From a product standpoint, the target retinoic acid will meet ≥98% chemical purity, as determined by HPLC, with confirmed structural identity by LC–MS. Product solubility and stability will be evaluated in representative cosmetic and pharmaceutical-relevant solvent systems, with a target of ≥90% chemical stability over 6 months under controlled storage conditions (4 °C and ambient temperature, protected from light). From a biomanufacturing perspective, the project targets β-carotene titers exceeding 20 g/L in fed-batch fermentation and quantitative (>90%) enzymatic conversion of intracellular β-carotene to retinoic acid, yielding final retinoic acid titers of ≥15–20 g/L directly from glucose. Additional performance metrics include volumetric productivities ≥0.1–0.2 g/L/h, fermentation reproducibility with batch-to-batch variability <10%, and downstream recovery yields exceeding 80% using scalable purification workflows. Collectively, these specifications are designed to enable cost-competitive, scalable, and regulatory-compliant production of retinoic acid suitable for cosmetic and pharmaceutical applications.			
R&D Plan	The overall research plan is designed to develop and validate a scalable, bio-based fermentation platform for the production of retinoic acid from glucose via high-titer β-carotene biosynthesis in *Yarrowia lipolytica*. The research approach, performance specifications, and milestones will be refined in collaboration with an industry partner to ensure alignment with commercial requirements and downstream application needs.			

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Stage 1: Strain Engineering and Pathway Integration (Month 0–9)

Objectives:
Establish robust recombinant *Y. lipolytica* strains capable of high-level β-carotene accumulation and efficient enzymatic conversion to retinoic acid.

Key Tasks and Methodologies:

- Validate baseline β-carotene production in previously engineered strains containing two copies of *ACC1/DGA*, single-copy overexpression of mevalonate pathway genes, and six copies of *GGPPsa*, *CarRP*, and *CarB*.
- Introduce and chromosomally integrate the retinoic acid biosynthetic module, including genes encoding β-carotene 15,15′-monooxygenase (BCMO1), retinol dehydrogenase, and retinaldehyde dehydrogenase.
- Optimize gene copy number, promoter strength, and genomic integration sites to maximize β-carotene conversion efficiency while maintaining host growth and redox balance.
- Perform shake-flask screening to quantify β-carotene, retinal, retinol, and retinoic acid levels using spectrophotometry, HPLC, and LC–MS.

Expected Milestones:

- Engineered strains exhibiting ≥20 g/L β-carotene accumulation.
- Demonstration of detectable retinoic acid production with ≥50% conversion of intracellular β-carotene.
- Selection of lead production strains for fermentation optimization.

Stage 2: Fermentation Optimization and Scale-Up (Month 9–18)

Objectives:
Maximize retinoic acid titers, productivity, and reproducibility under scalable fermentation conditions.

Key Tasks and Methodologies:

- Optimize batch and fed-batch fermentation parameters, including carbon feeding strategies, dissolved oxygen, pH, temperature, and C:N ratio, at bench scale (1–5 L).
- Evaluate the impact of lipid accumulation and intracellular product sequestration on retinoic acid formation and stability.
- Scale lead strains to pilot-scale bioreactors (up to 20–30 L) to assess robustness and batch-to-batch variability.
- Conduct preliminary technoeconomic analysis (TEA) to identify key cost drivers and guide further strain and process optimization.

Expected Milestones:

- Retinoic acid titers ≥15–20 g/L in fed-batch fermentation.
- Volumetric productivity ≥0.1–0.2 g/L/h with batch-to-batch variability <10%.
- Identification of commercially relevant operating windows for scale-up.

Stage 3: Downstream Processing and Product Validation (Month 18–24)

Objectives:

Develop a scalable downstream processing workflow that delivers high-purity retinoic acid suitable for cosmetic and pharmaceutical applications.

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Key Tasks and Methodologies:

- Optimize extraction and purification strategies to recover retinoic acid with minimal degradation, focusing on solvent selection, phase separation, and chromatography where necessary.
- Remove host cell contaminants and residual intermediates to achieve ≥98% product purity.
- Evaluate product stability under relevant storage conditions (light exposure, temperature, formulation-compatible solvents).
- Refine technoeconomic analysis incorporating downstream recovery yields, solvent recycling, and cost-of-goods estimates.
- Where possible, provide material samples to industry partners for preliminary application testing and feedback.

Expected Milestones:

- Downstream recovery yield ≥80% with ≥98% retinoic acid purity.
- Demonstrated product stability over ≥6 months under controlled storage conditions.
- Defined cost and performance benchmarks supporting commercial feasibility.

Overall Project Outcome

Successful completion of this research plan will deliver a fully integrated, fermentation-based platform for the sustainable production of retinoic acid from glucose. The project will de-risk key technical challenges spanning strain engineering, fermentation scale-up, and downstream processing, positioning the technology for rapid translation to industrial manufacturing through strategic industry partnerships.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Rensselaer Polytechnic Institute (RPI) research team, led by Prof. Mattheos Koffas, will have primary responsibility for executing all technical aspects of the proposed R&D project. The RPI team will design, construct, and evaluate recombinant *Yarrowia lipolytica* strains for high-titer β-carotene production and its enzymatic conversion to retinoic acid, in accordance with the R&D plan. This includes metabolic pathway design, gene synthesis and chromosomal integration, strain screening, and analytical validation of pathway performance. RPI will conduct fermentation development and optimization studies at bench scale and pilot scale (up to 30 L), including batch and fed-batch operation, media optimization, and process control. These activities will be supported by major infrastructure available through RPI's Microbiology and Fermentation Core Facility, which provides access to stirred-tank bioreactors, process control systems, and analytical instrumentation for microbial fermentation. Genetic construction and strain validation will be supported by RPI's Genomics Core Facility, including DNA synthesis, sequencing, and molecular biology services. Analytical characterization of β-carotene, retinoic acid, and pathway intermediates will be performed using spectrophotometric, HPLC, and LC–MS methods available through RPI's Analytical Biochemistry Core Facility. Additional laboratory space, equipment, and expertise for strain engineering, cultivation, and metabolite analysis are available within Prof. Koffas's research laboratories. If required to meet project milestones, specialized equipment for metabolite extraction, purification, and stabilization may be acquired through the project budget and installed at RPI. Collectively, these resources enable RPI to deliver all planned technical objectives, from strain construction through fermentation scale-up and product validation, while providing a robust foundation for collaboration with industry partners and future commercialization efforts.
Expected R&R of Korean SME Partner	The Korean SME will collaborate closely with the RPI research team to define commercially relevant target specifications, performance benchmarks, and technical milestones for bio-based retinoic acid production. Drawing on its market knowledge and downstream application expertise, the Korean SME will provide input on desired product attributes, quality requirements, and cost targets relevant to cosmetic, personal care, and biomedical applications. The Korean SME will conduct performance testing and validation of retinoic acid samples produced by the RPI team, with particular emphasis on evaluating stability, compatibility, and performance under real-world processing, storage, and formulation conditions. These evaluations may include testing under varying temperature, light exposure, and solvent or formulation environments representative of commercial use. Where appropriate, the Korean SME will facilitate preliminary customer evaluations to assess product acceptance and application performance.

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	In addition, the Korean SME will contribute to development planning by providing feedback that informs iterative optimization of strain performance, fermentation parameters, and downstream processing workflows. The SME will commit technical personnel, testing infrastructure, and formulation or application expertise to support these activities. Through ongoing coordination with RPI, the Korean SME will help ensure that the developed technology meets market needs and accelerates translation from laboratory-scale production to commercially viable manufacturing.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this joint R&D project include the development of a sustainable, fermentation-based platform for the production of retinoic acid from renewable carbon sources. Key technological achievements will include engineered *Yarrowia lipolytica* strains capable of high-titer β-carotene biosynthesis and its efficient enzymatic conversion to retinoic acid, as well as optimized fermentation and downstream processing workflows suitable for scale-up. Collectively, these advances will deliver a bio-based alternative to conventional chemically synthesized retinoic acid with reduced environmental impact and improved supply-chain resilience. From a commercialization perspective, the project will establish a manufacturable retinoic acid production process that meets industry-defined performance, purity, and cost specifications for cosmetic and pharmaceutical applications. By achieving integrated strain performance, fermentation robustness, and product quality validation in collaboration with an industry partner, the technology is expected to reach Technology Readiness Level (TRL) 7 by the end of the project. This level of maturity will provide a strong foundation for pilot manufacturing, regulatory engagement, and product development leading to commercial deployment. Broader impacts include enabling domestic production of a strategically important bioactive ingredient, reducing reliance on petrochemical and agriculturally constrained supply chains, and supporting the growth of sustainable biomanufacturing industries. Successful completion of this project will contribute to economic development by lowering barriers to commercialization of bio-based retinoids, fostering industry–academia collaboration, and strengthening the competitiveness of fermentation-derived specialty chemicals in global markets.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization